# 9.7 Equal Sums of Powers

WORKSHEET

**The question**

**STEP I 1992, Question 5**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q5

## Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

**Ask yourself** If a polynomial $\mathrm{p}(x) = \sum c_k x^k$ has $\mathrm{p}(1) = 0$, what does that say about the coefficients? What about $\mathrm{p}'(1) = 0$?

### STAGE 1 A triple root

**Ask yourself** Factorise $1 - x^2$ and $1 - x^4$. How many factors of $1 - x$ appear?

### STAGE 2 Derivatives at 1

**Ask yourself** If $(1-x)^3$ divides $\mathrm{p}_0$, what power of $(1-x)$ divides $\mathrm{p}_0'$? And then $\mathrm{p}_1 = x\mathrm{p}_0'$ and $\mathrm{p}_1'$?

### STAGE 3 Reading off the coefficients

**Ask yourself** What are $\mathrm{p}_0'(1)$ and $\mathrm{p}_1'(1)$ in terms of the coefficients $c_k$?

### STAGE 4 One more factor

**Pause and try** To make the cubes balance too, we need a fourth factor of $(1-x)$. Which polynomial should replace $\mathrm{p}_0$?

## Try this next

1 **Shifting.** (From STEP II, 2025.) Suppose $\sum a_k = \sum b_k$ and $\sum a_k^2 = \sum b_k^2$ for two lists of the same length $m$. Show that $\sum a_k^3 + \sum (c + b_k)^3 = \sum b_k^3 + \sum (c + a_k)^3$ for every $c$. Use this, starting from a balanced split of $\{1, \ldots, 8\}$, to find a split of $\{1, \ldots, 16\}$ in which the cubes balance as well.

2 **Squares of squares.** (Also from STEP II, 2025.) The sets $A = \{1, 3, 4, 5, 9, 11\}$ and $B = \{2, 6, 7, 8, 10\}$ have equal sums and equal sums of squares. Split $\{n^2, (n+1)^2, \ldots, (n+11)^2\}$ into two sets with equal sums, for every positive integer $n$. (*Hint:* add 0 to the second set.)

3 **Translation invariance.** Show that if two sets of numbers have equal sums of $r$th powers for $r = 0, 1, \ldots, k$, then so do the sets obtained by adding the same number $c$ to every element. Why does this mean that the question's sets $\{1, \ldots, 15\}$ could

equally be replaced by any 16 consecutive integers?

**4 Fair sweets.** Sweets have values $1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{8}$. Compare the totals $A$ gets with the orders $ABABABAB$ and $ABBABAAB$.

## Further reading


- **Jean-Paul Allouche & Jeffrey Shallit, "The ubiquitous Prouhet–Thue–Morse sequence"**, in *Sequences and Their Applications* (Springer, 1999). A survey of the sequence's many appearances; free online.
- **Matt Parker, *Things to Make and Do in the Fourth Dimension*** (Penguin). Includes equal power sums and other number curiosities.
- **Joshua Cooper & Aaron Dutle, "Greedy Galois games"**, *American Mathematical Monthly* 120 (2013). Fair turn-taking and Thue–Morse.


The full guided solution, and what the question leads to at university, is in Section 9.7:
https://beyondstep.keithmcnulty.org/ch09/equal-power-sums.html